Indian Institute of Management Rohtak

Information Technology Applications

Detecting Money Laundering Networks
(A machine learning workflow)

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1. Introduction

Money laundering is a process used to conceal the illegal origin of money gained by criminal activity. UN Office on Drugs and Crime estimates that money laundering results in laundered proceeds of around $800 billion to $2 trillion annually, which is around 2 to 5 percent of global GDP each year. All banks and other financial institutions are mandated under anti-money laundering (AML) legislation to carry out various levels of transaction monitoring and to report on certain types of suspicious activity. This is a costly requirement, and to help minimize the costs associated with compliance, these checks are often carried out using automated processes.

Why rule-based systems are not enough. The most common approach to building a machine learning system for fighting money laundering is to automate the process of reviewing transactions using automated rules. Often, this might start with basic rules, such as: Transactions over $10,000 must be reported. Transfers sent to more than 5 recipients in a day must be reported. While rules such as these might catch many mundane money laundering attempts, they are well known to money launderers. Any money laundering that can be easily detected by rules will be avoided in favour of more complex methods that make it harder to generate such rules. Rules are easy to understand, and money launderers take the time to ensure their methods do not trigger these rules. In contrast, a machine learning system trains and tests models on past transactions it has been trained to identify as money laundering versus legitimate, and automatically applies what it has learned to future transactions. Instead of applying simple rules that everyone is aware of, it learns to detect a wide range of patterns of behaviour in transactions that have been identified as money laundering. Rather than just triggering on some set of predetermined characteristics, the system is able to classify transactions into the types of money laundering or legitimate activity it has been trained on.

2. Dataset and Exploratory Analysis

After extraction from a compressed archive, the dataset was loaded with no missing values across any of its eight columns.

Column	Type	Description
TX_ID	int64	Unique transaction identifier
SENDER / RECEIVER ACCOUNT_ID	int64	Originating and destination accounts
TX_TYPE	object	Transaction category (e.g. TRANSFER)
TX_AMOUNT	float64	Monetary value of the transaction
TIMESTAMP	int64	Sequential step ID, range 0 to 199
IS_FRAUD	bool	Ground-truth binary fraud label
ALERT_ID	int64	Alert reference; -1 for non-alerted rows

A key finding from EDA is that fraudulent transactions carry very small amounts. The fraud median is $10.60 versus $156.97 for legitimate transactions, and zero percent of fraud cases exceed the 99th percentile of normal amounts. This is a structural property of the synthetic generator, not model leakage,

confirmed later by cross-validation. The TIMESTAMP column has a correlation of 0.0004 with the fraud label and was excluded as a non-informative feature. ALERT_ID directly encodes the label and was dropped before all modelling to prevent leakage.

3. Preprocessing and Feature Engineering.

The split-first pipeline was used as a strict method to remove data leakage. The entire setup was: drop identifier and label-correlated column, separate the target, one-hot encode TXTYPE, train-test split (stratified 80/20), and then compute account-level aggregate statistics only on training data, map those statistics to the test set using training-generated values with global medians as fallbacks, and finally, fit StandardScaler on training data only, then transform both sets.

There were 6 behavioural features that were developed: the frequency of sender and receiver transactions, the average transaction amount of sender and receiver, as well as the deviation of sender and receiver transactions. These records per-account behavioural patterns without recourse to any information through a test-set. There were two versions of the features (Version A featuring all seven features, including TXAMOUNT, and Version B featuring the four behavioural features only, without TXAMOUNT and deviation columns) to allow realistic model assessment not dependent on the trivially discriminating amount signal.

4. Building of Transactions Network.

NetworkX was used to construct a directed graph that contained 9,999 account nodes and 68,947 transaction edges. All accounts were measured using four per-node metrics: in-degree and out-degree based on transaction volume, betweenness centrality (estimated with k=500Sampling) to determine bridge accounts that are involved in transactions crossing the network, and the clustering coefficient to find accounts that were part of closed transactions within the network. These measures were kept in different places and were utilized in the detection of suspicious accounts. Their presence in the supervised feature matrix was avoided in order to prevent the leakage of graph-level features of the entire network to the test set.

5. Unsupervised Detection of Anomalies.

The scaled training set was furnished with an isolation Forest, with contamination of 0.05. Training and test scores were created on the basis of anomaly scores. The training scores threshold of the fifth percentile identified 52,930 miscellaneous transactions. K-Means five clusters were also fitted on training data, with cluster labels and centroid distance being projected on either set. The visualisation of PCA proved that there was definite structural separation of cluster grouping, with one cluster having the concentration of the anomalous transaction region.

A dual-criterion filter was then used to find suspicious accounts: minimum Isolation Forest score less than the fifth-percentile threshold and betweenness centrality greater than the 75th percentile. This outlier behaviour and high network influence identified 2349 accounts out of 9999, which is 23.5 percent of the network.

6. Managed Classification and Assessment.

6.1 Training Strategy and Models.

Three classifiers were trained using leakage-free preprocessed data. Random Forest set maxdepth=8 and minsamplesleaf=10, with classweight=balanced, which does not allow any unlimited growth of the trees and reweights the minor 0.13 percent group. Since KNN does not provide the ability to weight classes, KNN was trained on a balanced undersample of 13,750 majority-class and 1,375 minority-class instances with a 9.1 percent training fraud rate. classweight balanced with regularisation C=.1 Logistic Regression.

6.2 Resolving the RF = 1.0 Finding

The first Random Forest generated a perfect ROC-AUC of 1.0000. This is not a leakage but a dataset property that was determined on a systematic diagnosis. The design of TX_AMOUNT is almost a perfect discriminator, as it is designed so that zero percent of fraud transactions are in the 99 th percentile of legitimate transactions. The correlation of TIMESTAMP between the two columns was 0.0004, eliminating leakage in that column. Fold scores of 0.9998, 0.9996, 0.9989, 0.9998, 0.9995 with a standard deviation of 0.0003 were obtained by a five-fold stratified cross-validation which was run with a depth-3 tree, indicating that the high performance was inherent in the data, not due to memorisation.

6.3 Results

Model	Features	ROC-AUC	Fraud Recall	Fraud Precision
Random Forest	All (Version A)	0.9999	95.9%	93.8%
Random Forest	Behavioural (B)	0.7941	55.5%	1.0%
KNN	Behavioural (B)	0.7074	18.0%	0.8%
Logistic Regression	Behavioural (B)	0.7265	62.2%	0.3%

The behavioural Random Forest achieved a ROC-AUC of 0.7941, within the 0.75 to 0.90 range typical of real-world AML models. Logistic Regression led on fraud recall at 62.2 percent, identifying the highest proportion of true fraud cases at the cost of precision. Feature importance analysis on the full-feature model showed sender_tx_deviation driving 50.6 percent of decisions and TX_AMOUNT contributing 32.0 percent, together accounting for 82.6 percent of the model's classification logic.

7. Final Dashboard

The dashboard below consolidates all key pipeline outputs into a single visualisation. All values are drawn directly from confirmed notebook execution results.

The dashboard is displayed in a 5 horizontal row layout. On Row 1, there are five KPI tiles: total transactions (1,323,234), confirmed fraud (1,719 at 0.1299%), Isolation Forest flagged transactions (52,930), suspicious accounts (2,349), and network dimensions (9,999 nodes, 68,947 edges). Row 2 indicates the clustered ROC-AUC and fraud recall comparison in all four model specifications, the

feature importance bar chart with the dominant sender tx deviation of 50.6%, and the five-fold cross-validation scores verifying the standard deviation of only 0.0003. Row 3 shows the log-scale distribution of TX-AMOUNT between fraud and non-fraud amount, the colour-coded confusion matrix of the behavioural Random Forest, and a radar chart showing all the models by ROC-AUC, non-fraud recall, precision, PR-AUC, and non-fraud stability. Row 4 indicates the Isolation Forest anomaly score histogram with the fifth-percentile threshold highlighted, the seven-stage pipeline architecture flow chart, and the precision-versus-recall tradeoff chart. The last scorecard that verifies the pipeline integrity and the leakage verdict is in Row 5, the fraud proportion donut, and the network centrality degree distribution.

Figure 1: AML Intelligence System: Final Detection Dashboard

8. Conclusion

The AML detection pipeline was able to process 1,323,234 transactions with a total of 52,930 transactions being recognized as anomalous and 2,349 suspicious accounts being detected using the network centrality and the Isolation Forest requirements. The distributional analysis and cross-validation of the systematic study of the RF = 1.0 result verified that the perfect separation is an artifact of the synthetic generator and not some defect of the pipeline. The behavioural model, which has an ROC-AUC of 0.7941, was in line with published real-world AML benchmarks. Integrating the unmonitored anomaly detection with the supervised classification offered a multi-layered detection system, which could bring out the familiar and unseen patterns of fraud, in addition to which the Logistic Regression offered the highest recall of fraud at 62.2 percent with an arguably difficult set of behavioural features.

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